

Python Basics - Study Document

1. Introduction

Python is a high-level, interpreted programming language. It is easy to learn and is widely used for web development, data analysis, artificial intelligence, automation, and education.

2. Variables

A variable is a name used to store a value. Python does not require a separate declaration of the variable type.

3. Data Types

Common Python data types include int, float, string, boolean, list, tuple, set, and dictionary.

4. Conditional Statements

Conditional statements are used to make decisions in a program. The main keywords are if, elif, and else.

5. Loops

Loops are used to execute a block of code repeatedly. Python provides for and while loops.

6. Functions

A function is a reusable block of code that performs a specific task. Functions are defined using the def keyword.

7. Advantages of Python

Python has simple syntax, a large standard library, strong community support, and can be used for many types of applications.

8. Conclusion

Python is a beginner-friendly programming language with applications in many areas of software development and technology.